

Saad Haider

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Github | Website | LinkedIn

EDUCATION

Indian Institute of Technology, Jammu B.Tech.	2024 – 2028 CGPA: 8.11
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PROJECTS

• Sitecast - Automated UI Demo Generator <i>Playwright, Redis, FFmpeg</i> – Built a distributed video generation system for automated UI demonstrations. – Architected a queue-driven worker system using BullMQ, Redis, and Docker. – Engineered a Playwright + Xvfb + FFmpeg recording pipeline for headless video generation. – Developed a human-like scrolling engine with adaptive reading pauses and smooth scrolling.	<i>June 2026</i> GitHub
• Anton - SQL Database Engine <i>Java</i> – Built a SQL database engine with custom query parsing, execution, and persistent storage. – Implemented a B+ Tree indexing engine supporting point lookups and range queries. – Designed page management, record serialization, and catalog metadata layers. – Developed a SQL execution engine supporting DDL, CRUD, and WHERE query execution.	<i>Oct. 2025</i> GitHub
• Ember - Redis-compatible In-Memory KV Store <i>Rust, Tokio, TCP Sockets</i> – Built a Redis-compatible in-memory KV store with RESP support over raw TCP sockets. – Engineered a multi-core concurrent architecture achieving ~200k requests/sec in local benchmarks. – Implemented crash-safe RDB snapshot persistence with atomic file swaps and recovery. – Designed a hybrid TTL expiration engine with request pipelining and atomic transactions.	<i>Mar. 2026</i> GitHub
• Orin - Programming Language Interpreter <i>Java</i> – Built a tree-walk interpreter with lexical analysis, parsing, AST generation, and evaluation. – Implemented lexical scoping, nested environments, first-class functions, and closures. – Implemented classes, methods, object instantiation, and dynamic dispatch. – Engineered a modular Visitor-based AST architecture for extensible language features.	<i>Oct. 2025</i> GitHub

OPEN SOURCE CONTRIBUTIONS

[VIEW CONTRIBUTIONS](#)

- Implemented new static-analysis checks and configurable rules in **Checkstyle**, a Java code-quality tool.
- Implemented new modules in **Checkstyle**.
- Added Netlify adapter and Nuxt adaptors to **Dualmark** for framework-agnostic Markdown delivery.

TECHNICAL SKILLS

- **Languages:** Java, Rust, TypeScript, JavaScript
- **Frameworks:** React, Next.js, React Native, Express.js
- **Tools & Systems:** Linux, TCP Sockets, Redis, PostgreSQL, Docker

ACHIEVEMENTS

• Secured 4th place in Hack-O-Calypse, a competitive programming hackathon at IIT Jammu.	<i>2024</i>
• Secured 2nd place in Developer of the Month, a web development hackathon at IIT Jammu.	<i>2024</i>

POSITIONS OF RESPONSIBILITY

• Web Development Mentor , Coding Club, IIT Jammu	<i>Present</i>
• Competitive Programming Mentor , Coding Club, IIT Jammu	<i>Present</i>